Course Policy
Statistical Structures in Data and Inference

Mukesh Patel School of Technology Management and Engineering
Data Science Department

Course Policy

Program/Branch/Semester	:	MBATech/Data Science/IV
Academic Year	:	2025-26
Course Code & Name	:	702DB0C006 & Statistical Structures in Data and Inference
Credit Details	:	L T P C 3 0 2 4
Course Coordinator Faculty	:	Dr. Siba prasada Panda
Contact No. & Email	:	8080616910 siba.panda@nmims.edu
Office	:	MPSTME New Building, Department Office, 6th floor
Office hours	:	
Other Course Faculty members teaching this course	:	Course Faculty 1: Dr. Manishkumar S. Tiwari Contact No. & Email: 9022698001 & Manishkumar.tiwari@nmims.edu Office: Office Hours:
Queries by Emails are encouraged.		
Course link	:	Web Portal Link

1. Introduction to the Course

1.1. Importance of the course

"Statistical Structures in Data and Inference" is a crucial subject in our data-driven world. It equips us with the tools to uncover hidden patterns, quantify uncertainty, and draw valid conclusions from data. This allows us to make informed decisions, predict future behaviour, and drive innovation across various fields, from science and business to social sciences and artificial intelligence. By mastering statistical structures, we develop critical thinking skills and learn to analyse evidence effectively, solve problems creatively, and communicate complex information clearly. In short, understanding statistical structures empowers us to navigate the data deluge and thrive in the 21st century.

1.2. Objective of the Course

The subject "Statistical Structures in Data and Inference" equips students with the fundamental knowledge and tools to effectively analyze and interpret data, a crucial skill in our data-driven world.

Its primary objective is to:

1. Enable students to understand the underlying statistical structures present in data sets. This includes identifying patterns, trends, and relationships between variables. By leveraging statistical models and techniques, students can extract meaningful insights from the data, even if it is noisy or incomplete.

2. Provide students with the ability to draw valid conclusions and make informed decisions based on statistical evidence. This involves understanding the concepts of probability, hypothesis testing, and statistical inference. By applying these principles, students can quantify uncertainty, assess the significance of results, and make confident predictions about future outcomes.

3. Cultivate strong critical thinking and problem-solving skills. Through the process of analyzing data, students learn to identify biases, evaluate assumptions, and formulate logical arguments. This skillset is essential for success in various fields, ranging from research and academia to business and government.

4. Equips students with the communication skills necessary to effectively convey complex statistical information. This includes constructing clear and concise visualizations, interpreting statistical results, and presenting findings to diverse audiences.

1.3. Pre-requisite

Managing Uncertainty (SEM - III)

2. Course Outcomes (CO) and mapping with Program Outcomes (PO)

2.1. Course Outcomes

Course outcomes:

After completion of the course, students will be able to

CO1: Recall fundamental concepts of confidence intervals, hypothesis testing, and maximum likelihood estimation (MLE) (L1- Remembering)

CO2: Explain the likelihood function, maximum likelihood estimation (MLE) across various distributions (Binomial, Multinomial, Gaussian), and apply the continuous version of Bayes' theorem. (L2 -Understanding and L3- Applying)

CO3: Perform hypothesis testing using Z-tests, t-tests, and Chi-square tests for independence and goodness of fit, including interpretation of p-values, Type I and Type II errors. (L3-Applying and L4 – Analyzing)

CO4: Interpret generalized linear models (GLMs) such as logistic regression for binary outcomes, including log-odds interpretation and model diagnostics. (L2-Understanding)
CO5: Perform and interpret Ridge and Lasso regression for feature selection with Python implementations and visualizations (L4-Analyzing)

CO6: Evaluate and create advanced statistical models, including logistic regression for binary outcomes and non-parametric tests (Sign Test, Rank Sum Test, Kruskal-Wallis, Friedman tests), with Python implementations and visualizations. (L5 – Evaluating and L6-Creating)

CO-PO mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	-	-	1	-	-	-	-	-	2	2	-
CO2	3	2	-	2	2	-	-	-	1	-	2	3	2
CO3	3	3	-	3	2	1	-	-	2	-	2	3	3
CO4	3	2	-	2	3	-	-	-	2	-	3	3	2
CO5	3	3	2	3	3	1	-	-	2	-	3	3	2
CO6	3	3	3	3	3	2	-	-	2	-	3	3	3

Mapping Levels: 3- High, 2-Medium, 1-Low

3. Syllabus, Pre-class activity and References

3.1. <u>Teaching and evaluation scheme</u>

Teaching Scheme				Evaluation Scheme	
Lecture Hours per week	Practical Hours per week	Tutorial Hours per week	Credit	Internal Continuous Assessment (ICA) As per Institute Norms (50 Marks)	Theory (3 Hrs, 100 Marks)
3	2	0	4	Marks Scaled to 50	Marks Scaled to 50

3.2. <u>Syllabus</u>

Unit	Description	Duration

1	**Statistical Inference** • Confidence intervals • Likelihood function and maximum likelihood • Computing the MLE • Computing the MLE: examples • Plotting the likelihood • Inference example: frequent • Inference example: Bayesian • Continuous version of Bayes' theorem • Posterior intervals	8
2	**Hypothesis Testing** • Basics of Estimating the populations mean and difference • Estimating the proportion and difference • Large sample test for population mean, difference • Large sample test for proportion, difference. • Sampling distribution of variance, estimation. • Inferences from small sample: Student's t distribution; Small sample t test for following – A population mean, A difference between two means, Confidence interval. • Rejection and Non-rejection region • Type I and Type II errors • testing hypothesis about a population mean using the Z- statistic • using p-values to test Hypothesis	10
3	**ANOVA/MANOVA** • Chi-Square as a test of independent • Chi-square as a Test of goodness of fit: Testing the Appropriateness of a Distribution, Analysis of Variance, •	3
4	**Regression Model and its Inference** • Least squares and linear regression: Introduction; Notation; Ordinary least squares; Regression to the mean; Linear regression; Residuals; Regression inference • Multivariable regression: Multivariate regression; Multivariate examples; Adjustment; Residual variation and diagnostics; Multiple variables, Interaction Terms, Non-linear Transformations of the Predictors, Qualitative Predictors • Multiple Regression Analysis: The Problem of Estimation and the Problem of Inference • Dummy Variable Regression Models • Multi-collinearity, Heteroscedasticity, Autocorrelation • Econometric Modelling: Model Specification and Diagnostic Testing • Correlation and Covariance Analysis • Canonical Analysis, Canonical Roots/variates	12

5	**Extension of regression analysis** • Ridge Regression and Lasso • Nonlinear Regression Models: Approaches to Estimating Nonlinear Regression models	4
6	**Generalized linear models** • Logistic Regression • Binary outcomes • Count outcomes • Multiple Logistic Regression	4
7	**Nonparametric Test** • Sign Test • Rank Sum Test • Kruskal-Wallis and Friedman tests	4
	Total hours	45

3.3. Pre-class activity

Outline for preliminary study to be done for each unit will be provided prior to commencement of each unit. Preliminary study material (video links, presentation, notes etc) will be made available on the student portal. Students are expected to go through this material before attending the upcoming session. It is expected that the students put in at least two hours of self-study for every one hour of classroom teaching. During the lecture session, more emphasis will be given on in-depth topics, practical applications and doubt solving.

3.4. References

Text Books

1. Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani, *An Introduction to Statistical learning with application in R*, Springer, 2013
2. Damodar Gujarati, Basic Econometrics, 5th edition, McGraw Hill Education, 2017
3. Richard Levin,David S Rubin, Sanjay Rastogi and Massod Husain, "Statistics For Management", Pearson, 8th edition, 2017

Reference Books

1. Agresti , *An Introduction to Categorical Data Analysis*, 3rd edition, Wiley, 2018
2. Trevor Hastie, Robert Tibshirani, and Jerome Friedman, The Element of Statistical Learning, Data mining, Inference and Prediction, Springer, 2013

3.5. Other resources:

Youtube Links:
https://nptel.ac.in/courses/111105091

https://www.youtube.com/watch?v=I_dhPETvll8

Note: The latest edition of books should be referred.

4. Laboratory details

Students are expected to recall the fundamental theory concepts relevant to the exercise to be performed in the upcoming laboratory.

The following 14 exercises and one design of lab problem statement will form the submission for laboratory coursework.

Sr. No.	Week No.#	List of Lab Exercises	Mapped CO
1	1	Probability calculation using Python • Binomial distribution • Normal distribution • Random number generation	CO2 and C03
2	2	Implementation of likelihood function and calculation of maximum likelihood in Python and Excel	CO2 and C03
3	3	• Bernoulli/binomial likelihood • Draw an Inference of Bayesian model through Log likelihood estimation • Case study: predict gender through height and weight.	CO3
4	4	Parametric Statistical Hypothesis Tests: How to Perform Parametric Statistical Hypothesis Tests in Python • Large sample test for population mean • Large sample test for proportion	CO3
5	5	Parametric Statistical Hypothesis Tests: How to Perform Parametric Statistical Hypothesis Tests in Python • Paired Student's t-Test- can be implemented in Python using the ttest_rel() SciPy function	CO3
6	6	One-Way ANOVA Versus Two-Way ANOVA Check Assumption for ANOVA	CO3

		Case Study-1: determine whether there are any statistically significant differences between the means of three or more independent (unrelated) groups. Case Study-2: compare worker productivity based on two independent variables, such as salary and skill set. It is utilized to observe the interaction between the two factors and tests the effect of two factors at the same time	
7	7	One-Way MANOVA Versus Two-Way MANOVA Check Assumption for MANOVA Case Study: test how drug metabolism is influenced by a malaria-like infection	CO2
8	8	Simple Linear Regression: Case study: Boston data set, which records medv (median house value) for 506 census tracts in Boston. • Predict medv using 12 predictors such as rm (average number of rooms per house), age (average age of houses), and lstat (percent of households with low socioeconomic status) • Predict medv using Interaction effect • Predict car sales using Qualitative Predictors	CO2 and CO3
9	9	Simple and Multiple Linear Regression: Case study: Boston data set, which records medv (median house value) for 506 census tracts in Boston. • Predict medv using 12 predictors such as rm (average number of rooms per house), age (average age of houses), and lstat (percent of households with low socioeconomic status) • Predict medv using Interaction effect • Predict car sales using Qualitative Predictors • Predict car sales with Non-linear Relationships	CO1 and C03
10	10	How to deal with Potential Problems of Regression Case study: Boston data set, which records medv (median house value) for 506 census tracts in Boston.	CO1
11	11	Linear Model Selection and Regularization: • Subset Selection • Shrinkage Methods Data Set: This dataset was originally taken from the StatLib library which is maintained at Carnegie Mellon	CO3

		University. This is part of the data that was used in the 1988 ASA Graphics Section Poster Session. The salary data were originally from Sports Illustrated, April 20, 1987. The 1986 and career statistics were obtained from the 1987 Baseball Encyclopedia Update published by Collier Books, Macmillan Publishing Company, New York. Case Study-1: apply the best subset selection approach to the Hitters data. We wish to predict a baseball player's Salary on the basis of various statistics associated with performance in the previous year Case Study-2: Will now perform ridge regression and the lasso in order to predict Salary on the Hitters data	
12	12	Logistic Regression: Data Set: This data set consists of percentage returns for the S&P 500 stock index over 1, 250 days, from the beginning of 2001 until the end of 2005 Case Study: fit a logistic regression model in order to predict Direction using Lag1 through Lag5 and Volume	C03
13	13	Cross-Validation: Data Set: This dataset was taken from the StatLib library which is maintained at Carnegie Mellon University. The dataset was used in the 1983 American Statistical Association Exposition. The original dataset has 397 observations, of which 5 have missing values for the variable "horsepower" Case Study: Explore the use of the validation set approach in order to estimate the test error rates that result from fitting various linear models on the Auto data set	C02
14	14	Nonparametric Test: Case Study-1: Tests whether the distributions of two independent samples are equal or not. Case Study-2: Tests whether the distributions of two paired samples are equal or not.	C03

5. Tutorial Plan

No Tutorial for this course

Sr. No.	Week No.#	Tutorial exercises / activity	Mapped CO

6. Assessment Policy

6.1. Component wise Continuous Evaluation Internal Continuous Assessment (ICA) and Term End Examination (TEE)

Assessment Component	ICA (50 Marks)					TEE (100 marks) (Marks scaled to 50)
	Regular Lab Performance (A)	Class participation (B)	Mini Project (C)	Class Test 1 and Class Test 2 (D)	Assignment (E)	
Marks	5	5	10	10+10	10	100
Date/week of activity	Every Weekly	Throughout the semester	Week 14 & 15	As scheduled in academic calendar	Throughout the semester	As scheduled in academic calendar

6.2. Assessment Policy for Internal Continuous Assessment (ICA)

A: Assessment based on regular individual student assessment during each laboratory session. Each experiment/programming exercise may be given 5/10 marks with breakup on involvement, performance, application and learning etc. Record of the same to be maintained on the same day, scaled down to the required at the end of the term and shared with the student. Demonstration of practicums (where applicable, to the class/batch). Changes may be made based on the teaching scheme (Practical/Tutorial or both).

B: For viva, group evaluation should be avoided. Questions should be asked to assess conceptual learning and application. At the end doubts should be cleared with unanswered questions explained.

C: The assessment under this component should be based on the learning of the student in and out of the classroom. Following assessment activities may be considered-

1. Challenging assignments to fast learners. Individual assignments should be given to students to the extent possible. Assignments to test the understanding should be designed for slow/average learners. The assignments should be short and intended to test the understanding of the concept.

2. Mini projects on advanced topics not included in the syllabus. They should not be mere presentations or collection of data.

Components	Research analysis	Proposed Model	Result	Report
Marks	5	5	5	5

3. Presentation on latest technical innovation world over/ current active research in India

Components	Key Findings	Application of theories/concept understanding	Presentation style	Research Analysis
Marks	5	5	5	5

4. Journal paper-based activity. Complexity of the same should be based on the study level of the students

Components	Key Findings	Proof of Model/Survey	Submitted	Published
Marks	5	5	5	5

5. Developing pedagogical tools for active learning (simulations, videos, new challenging experiments/create game based on learning a specific topic/assignment questions/ any other

6. Articles on the relevant syllabus topics on platforms like Geeks for Geeks, Analytics Vidhya etc

- Different assessment activities may be given to different students. It is not necessary to have multiple activities per student.
- For a group of 2-3, informative short presentations should be arranged in the class or lab/tutorial sessions and not for the faculty alone.
- Efforts of the students should be appreciated irrespective of the outcome, where necessary scope for improvement should be highlighted.
- Transparency in terms of expectations from the activity/evaluation of the same should be maintained from the beginning for each assessment component.
- Faculty members should be wary of critical/judgmental approaches towards students.

D: Test 1 & 2 - First question should be compulsory and should cover all topics marked for the test. Thereafter an option to attempt two questions out of the remaining three. The questions should have a balanced mix of questions to include relevant Bloom's taxonomy levels of learning.

E: Class activity to assess the involvement and understanding of the student in the classroom (viz; think pair share/ games to test understanding/application of the topic/ rewarding challenging questions and application of mind/ any other).

Participation of students in national/international competitions, coding competitions, SRG projects, publications, startup initiatives, any other activity which adds to the learning of the student may also be considered in lieu of class interaction.

Components	First Case study Discussion	Second Case Study Discussion	Participation in Conference
Marks	2	3	5

6.3. Assessment Policy for Term End Examination (TEE)

A written(practical-based) examination of 100 marks for 3hrs duration will be conducted for the course as per the academic calendar.

7. Lesson Plan

Session No.	Topics	Mapped CO	Reference
1	**Statistical Inference** • Confidence intervals	CO1	TB1
2	• Likelihood function and maximum likelihood	CO1	TB1
3	• Computing the MLE	CO1	TB1
4	• Computing the MLE: examples	CO1	TB1
5	• Plotting the likelihood	CO1	TB1
6	• Inference example: frequent	CO1	TB1
7	• Inference example: Bayesian	CO1	TB1
8	• Continuous version of Bayes' theorem • Posterior intervals	CO1	TB1

9	**Hypothesis Testing** • Basics of Estimating the populations mean and difference	CO1	TB3
10	• Estimating the proportion and difference	CO1	TB3
11	• Large sample test for population mean, difference	CO1	TB3
12	• Large sample test for proportion, difference.	CO1	TB3
13	• Sampling distribution of variance, estimation.	CO1	TB3
14	• Inferences from small sample: Student's t distribution; Small sample t test for following – A population mean, A difference between two means, Confidence interval.	CO1	TB3
15	• Rejection and non-rejection region	CO1	TB3
16	• Type I and Type II errors	CO1	TB3
17	• testing hypothesis about a population mean using the Z- statistic	CO1	TB3
18	• using p-values to test Hypothesis	CO1	TB3
19	**ANOVA/MANOVA** • Chi-Square as a test of independent	CO1	TB3
20	• Chi-square as a Test of goodness of fit: Testing the Appropriateness of a Distribution, Analysis of Variance,	CO1	TB3
21	• Multivariate analysis of variance	CO1	TB3
22	**Regression Model and its Inference** • Least squares and linear regression: Introduction; Notation	CO3	TB1
23	• Ordinary least squares; Regression to the mean	CO3	TB1
24	• Linear regression; Residuals; Regression inference	CO3	TB1

25	• Multivariable regression: Multivariate regression; Multivariate examples	CO3	TB1
26	• Adjustment; Residual variation and diagnostics; Multiple variables, Interaction Terms	CO3	TB1
27	• Non-linear Transformations of the Predictors, Qualitative Predictors	CO3	TB1
28	• Multiple Regression Analysis: The Problem of Estimation and the Problem of Inference	CO3	TB1
29	• Dummy Variable Regression Models	CO3	TB1
30	• Multi-collinearity, Heteroscedasticity, Autocorrelation	CO3	TB1
31	• Econometric Modelling: Model Specification and Diagnostic Testing	CO3	TB1
32	• Correlation and Covariance Analysis	CO3	TB1
33	• Canonical Analysis, Canonical Roots/variates	CO3	TB1
34	**Extension of regression analysis** • Ridge Regression	CO3	TB1
35	• Lasso Regression	CO3	TB1
36	• Nonlinear Regression Models	CO3	TB1
37	• Approaches to Estimating Nonlinear Regression models	CO3	TB1
38	**Generalized linear models** • Logistic Regression	CO3	TB1
39	• Binary outcomes	CO3	TB1
40	• Count outcomes	CO3	TB1
41	• Multiple Logistic Regression	CO3	TB1

42	**Nonparametric Test** • Sign Test	CO2	TB2 & 3
43	• Rank Sum Test	CO2	TB2 & 3
44	• Kruskal-Wallis test	CO2	TB2 & 3
45	• Friedman test	CO2	TB2 & 3

8. Teaching-learning methodology

Faculty will make a group of 2-3 students for any group-based activity such as class participation, project, presentation etc. Lecture and laboratory session will be conducted as follows-

1. Lectures:

○ Outline for preliminary study to be done for each unit will be provided prior to commencement of each unit.

○ Deeper concepts and applications will be explained through Presentation and Video Lectures.

○ Real life-based case studies will be solved during the session on smart board or MS OneNote.

○ Some practical applications will be simulated in class on an online oracle platform for better understanding of the concepts, which will be available on GitHub for future reference.

2. Laboratory:

○ Lab manual consisting of theory and algorithms to support the lab experiment will be uploaded on the student portal.

○ Regular lab assessment and grading will be done. Students will be marked based on parameters like completion of lab assignment, originality, logic developed, interaction during the lab, submission, punctuality and discipline

9. Active learning techniques

Active learning is a method of learning in which students are actively or experientially involved in the learning process. Following active learning techniques will be adopted for the course.

1. Muddiest topic: Faculty will find out the least understood point/topic in the session. This topic is then further explained to ensure that it is understood well.

2. The "One Minute Paper": The faculty will ask students to take out a blank sheet of paper, pose a question (either specific or open-ended), and give them one (or perhaps two - but not many) minute(s) to respond.

3. Wait Time: Rather than choosing the student who will answer the question presented, this variation has the faculty WAITING before calling on someone to answer it.

4. Blended Learning: Students will be introduced to the topic at home while the in-depth topics, applications and numerical problems will be discussed by the faculty in the lecture session. Outline for preliminary study to be done for each unit will be provided prior to commencement of each unit. Preliminary study material (video links, presentation, notes etc) will be made available on the student portal.

5. Frame a question: Students will be asked to design and frame their own questions pertaining to the topic being taught. The idea is to stimulate students' curiosity, engage the students in collaborative teaching and learning, and motivate students to develop deeper understanding of the topic.

○ Frame questions for each unit of the course: At the beginning of each using, the faculty will create a new page in OneNote Class Notebook in collaborative section where every student will post his/her question.

○ Frame a question in lab: As discussed in section 6.2, student will be asked to design one unique lab problem based on the course syllabus.

6. Brainstorming: Students will be asked to generate ideas on a certain topic, category or question while the faculty will facilitate and record the answers on the blackboard/whiteboard.

10. Course Material

Following course material is uploaded on the student portal: <u>(give student portal link)</u>

- Course Policy
- Lecture Notes
- Lecture Videos
- Lecture Presentations
- Books / Reference Books / NPTEL video lectures link
- Assignments
- Lab Manuals, Test database link
- List of Program Outcomes

11. Course Outcome Attainment

Following means will be used to assess attainment of course learning outcomes.

- Use of formal evaluation components of continuous evaluation, assignments, laboratory work, semester end examination
- Informal feedback during course conduction

12. Academic Integrity Statement

Students are expected to carry out assigned work under Internal Continuous Assessment (ICA) independently. Copying in any form is not acceptable and will invite strict disciplinary action. Evaluation of corresponding components will be affected proportionately in such cases. Plagiarism detection software will be used to check plagiarism wherever applicable. Academic integrity is expected from students in all components of course assessment.